

ANALYSIS ON RANDOM FRACTALS

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1. Introduction

Almost all quantities defined here will depend on the choice of a fixed family of IFSs $\{F^\lambda : \lambda \in \Lambda\}$ and a fixed labelled tree T , see Section 2. This dependence will not normally be made explicit.

Later we will keep $\{F^\lambda : \lambda \in \Lambda\}$ fixed but consider different labelled trees $T = T(\omega)$ where $\omega \in \Omega$. Quantities that depend on the choice of T will be shown as functions of ω . Frequently, but not always, there will be a probability structure $(\Omega, \mathcal{F}, P)$ on Ω .

Most of this consists of extending parts of [Bar98, Section 5–7, pp 60–105] to the random setting. See also [Ham97].

2. Notation

2.1. Trees. A *tree* T is a finitely branching rooted tree, all of whose branches have infinite length.

The branching number at the node $w \in T$ is denoted by $U(w)$ and the edges out of w are denoted by $1, 2, \dots, U(w)$. See Figure 2.

In this way each $w \in T$ at level $|w| = n$ can be considered as a word $w = w_1 \dots w_n \in \mathbb{N}^n$. The root node is at level 0 and corresponds to the empty word $\emptyset$. Moreover, $1 \leq w_1 \leq U(\emptyset)$, and $1 \leq w_{i+1} \leq U(w_1 \dots w_i)$ for $1 \leq i \leq n-1$.

Let the set of words in T of length n be $T_n = \{w \in T : |w| = n\}$.

A *labelled tree* is a tree T together with a label function $V : T \rightarrow \Lambda$, where Λ indexes some family of IFSs:

$$(1) \quad \{F^\lambda : \lambda \in \Lambda\}, \quad F^\lambda = (\psi_1^\lambda, \dots, \psi_{M_\lambda}^\lambda).$$

If $V(w) = \lambda$ then the branching number $U(w)$ must equal M_λ .

The corresponding labelled tree is denoted by $(V(w); w \in T)$, or simply by T .

There is usually one or more underlying probability spaces $(\Omega, \mathcal{F}, P)$ associated with the space of labelled trees. A tree-valued random tree is denoted by T or $T(\omega)$. If we need to distinguish between different labelled trees, even in the deterministic case where there may be no underlying probability distribution, we again use the notation $T(\omega)$.

2.2. Coding maps, points and sets. For $w \in T_n$ define

$$(2) \quad \psi_w = \psi_{w_1}^{V(\emptyset)} \circ \psi_{w_2}^{V(w_1)} \circ \dots \circ \psi_{w_n}^{V(w_1 \dots w_{n-1})}.$$

For $\lambda \in \Lambda$ and $w = w_1 \dots w_n$ with $w_i \in \{1, \dots, M_\lambda\}$ define

$$\psi_w^\lambda = \psi_{w_1}^\lambda \circ \psi_{w_2}^\lambda \circ \dots \circ \psi_{w_n}^\lambda.$$

Each infinite branch of T corresponds to an *infinite* word w . We write $w \in T_\infty$ in this case. In particular $T_\infty \subset \mathbb{N}^\mathbb{N}$.

The truncation of a word w (finite or infinite) to its initial segment of length n is denoted by $w|n = w_1 \dots w_n$.

If $w \in T_\infty$ and $x_0 \in \mathbb{R}^n$ then under a uniform contractivity condition on the F^λ there exists $x_w \in \mathbb{R}^n$ independent of x_0 such that

$$(3) \quad \psi_{w|n}(x_0) \rightarrow x_w := \pi(w) \text{ as } n \rightarrow \infty.$$

Then

$$(4) \quad \pi(iw) = \psi_i(\pi(w)), \quad \pi(vw) = \psi_v(\pi(w)).$$

For $E \subset \mathbb{R}^n$ define

$$(5) \quad E_w = \psi_w(E).$$

The (compact) *fractal set generated by T* is

$$(6) \quad K = \{x_w : w \in T^\infty\} = \pi(T^\infty) = \lim_{n \rightarrow \infty} \bigcup_{w \in T_n} E_w$$

for any compact set E , where the limit is in the Hausdorff metric sense.

3. Uniformly pcf fractals

For notational simplicity we consider a family of contractive IFSs $\{F^\lambda = (\psi_1^\lambda, \dots, \psi_{M_\lambda}^\lambda); \lambda \in \Lambda\}$ acting on some compact subset of $\mathbb{R}^d$. However, since there is no geometry involved, it is not necessary to work in Euclidean space when working with pcf fractals as in this section, or with electric networks as in Section 7.¹ And in some ways it is even misleading to do so.

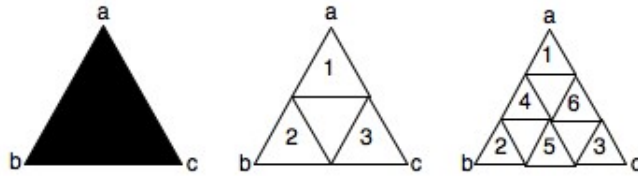


FIGURE 1. The set E and prefractals for $SG(2)$ and $SG(3)$.

3.1. Definitions. For each $\lambda \in \Lambda$ let K^λ be the unique compact attractor of F^λ . We say $\{F^\lambda : \lambda \in \Lambda\}$ is *uniformly pcf* if²

¹In fact it is not even necessary to work in a metric space at all, one can work in a suitable abstract symbolic framework. See [Kig01].

²Strictly speaking, the definition refers to the set of IFSs. One could also say that the fractal generated by every labelled tree with labels from $\{F^\lambda : \lambda \in \Lambda\}$ is a pcf fractal.

4.2. Properties.

Proposition 4.1. *Suppose $\{F^\lambda : \lambda \in \Lambda\}$ is uniformly pcf and T is a labelled tree with labels in Λ .*

- (1) *If $x \in V^{(n)}$ then $x = \psi_w(y)$ for one or more pairs $y \in V^{(0)}$ and $w \in T_n$.*
- (2) $V^{(n)} = \bigcup_{w \in T_n} V_w^{(0)}$.
- (3) *If $w \neq v \in T_n$ then $K_w \cap K_v = V_w^{(0)} \cap V_v^{(0)}$.*
- (4) $\pi^{-1} \circ \pi(P^{(n)}) = \pi^{-1}V^{(n)} = P^{(n)}$.
- (5) $V^{(0)} \subset V^{(1)} \subset V^{(2)} \subset \dots$.

5. Uniformly Nested and Affine Nested Fractals

5.1. Definitions. The family of IFSs $\{F^\lambda : \lambda \in \Lambda\}$ is *uniformly affine nested* if

- (1) The family is uniformly pcf and all F^λ act on $\mathbb{R}^n$.
- (2) The F^λ are affine nested fractals as in [Bar98, p69]³ with the additional constraint that the open set corresponding to the open set condition is independent of λ
- (3) *The set of essential fixed points⁴ for each $\lambda \in \Lambda$ is independent of λ and equals $V^{(0)}$.*⁵

The corresponding fractals K_w for each $w \in T$ are said to be *affine nested*.

The family of IFSs $\{F^\lambda : \lambda \in \Lambda\}$ is *uniformly nested* if it is uniformly affine nested and if for each λ the maps ψ_m^λ all have the same contraction factor α^λ .

5.2. Properties. Suppose $\{F^\lambda : \lambda \in \Lambda\}$ is uniformly affine nested. Then

- (1) $V^{(0)}$ is the set of essential fixed points.
- (2) For each λ , and independently of λ ,

$$P = \{m : \text{the fixed point of } \psi_m^\lambda \text{ is essential}\}.$$

- (3) Each essential fixed point is in exactly one n -complex for each $n \geq 1$.
- (4) Each 1-complex contains at most one essential fixed point.

6. Measures on Uniformly pcf Fractals

Suppose $\{F^\lambda = (\psi_1^\lambda, \dots, \psi_{M_\lambda}^\lambda) : \lambda \in \Lambda\}$ is uniformly pcf. Consider a fixed labelled tree $T = (V, T)$.

6.1. Definitions. Suppose $\theta^\lambda = (\theta_1^\lambda, \dots, \theta_{M_\lambda}^\lambda)$ satisfies

$$(10) \quad \sum_{m=1}^{M_\lambda} \theta_m^\lambda = 1, \quad 1 < \theta_m^\lambda < 1, \quad \lambda \in \Lambda.$$

Let

$$(11) \quad \theta_w = \theta_{w_1}^{V(\emptyset)} \theta_{w_2}^{V(w_1)} \dots \theta_{w_n}^{V(w_1 \dots w_{n-1})} \quad \text{for } |w| = n.$$

Example 6.1. In Figure 2 one has possible weights $(\theta_1^2, \theta_2^2, \theta_3^2)$ and $(\theta_1^3, \dots, \theta_6^3)$. Associated to the three lowest edges in T are the weights $\theta_1^2, \theta_2^2, \theta_3^2$. Associated to the 12 edges at the next level are weights $\theta_1^2, \theta_2^2, \theta_3^2, \theta_1^3, \dots, \theta_6^3, \theta_1^2, \theta_2^2, \theta_3^2$ respectively. Associated to each $w \in T_n$ is the weight obtained by multiplying the weights of the n edges in w .

³The ψ_m^λ are similitudes; they satisfy the open set condition; any two 1-cells are connected by a sequence of 1-cells (*connectivity*); reflection in the hyperplane bisecting the line segment joining any 2 points in the 0-cell maps n -cells to n -cells (*symmetry*); if $v \neq w \in T_n$ then $K_v \cap K_w = V_v^{(0)} \cap V_w^{(0)}$ (*nesting*).

⁴An essential fixed point is a fixed point x such that for some $m \neq n$ and for some fixed point y , $\psi_m^\lambda(x) = \psi_n^\lambda(y)$.

⁵This is shown to be a consequence in [Bar98, p 70, Prop 5.24].

Example 6.2. Suppose $\{F^\lambda : \lambda \in \Lambda\}$ is uniformly affine nested with the similitudes ψ_m^λ having contraction ratios α_m^λ . Then the natural weights are

$$\theta_m^\lambda = (\alpha_m^\lambda)^\beta, \quad \text{where} \quad \sum_{m=1}^{M_\lambda} (\alpha_m^\lambda)^\beta = 1.$$

For the uniformly nested case where each ψ_m^λ has contraction ratio α^λ independent of m , it is natural to take $\theta_m^\lambda = (M_\lambda)^{-1}$.

Define $\tilde{\mu} = \tilde{\mu}_\theta$ as the natural product measure on T_∞ , given on cylinder sets by

$$\tilde{\mu}_\theta\{v \in T_\infty : v|n = w\} = \theta_w.$$

Define $\mu = \mu_\theta$ on the fractal K corresponding to T by

$$\mu = \mu_\theta = \pi_\# \tilde{\mu}.$$

It is defined on the collection of Borel sets $\mathcal{B}(K)$, which is generated by the cylinder sets K_w for $w \in T$, i.e.

$$\mu_\theta(K_w) = \theta_w.$$

6.2. Properties. Essentially from the definitions

$$(12) \quad \int_K f d\mu = \sum_{w \in T_n} \theta_w \int f \circ \psi_w d\mu, \quad \mu = \sum_{w \in T_n} \theta_w \psi_{w\#} \mu,$$

where $\psi_{w\#} \mu$ is the pushforward measure given by $\psi_{w\#} \mu(E) = \mu(\psi_w^{-1} E)$.

We define μ_n to be the measure on $V^{(n)}$ obtained by pushing the μ -mass on each n -complex K_w onto its boundary $V_w^{(0)}$.

Let $N_0 = \#V^{(0)}$, the number of vertices in $V^{(0)}$. For $x \in V^{(n)}$ define

$$(13) \quad \mu_n(x) = \frac{1}{N_0} \sum_{w \in T_n} \theta_w \mathcal{X}_{V_w^{(0)}}(x).$$

That is, $x \in V^{(n)}$ obtains a contribution θ_w/N_0 from each n -complex $V_w^{(0)}$ to which it belongs.

Then

$$(14) \quad \mu_n \rightharpoonup \mu,$$

since for continuous f on K ,

$$\begin{aligned} \left| \int f d\mu - \int f d\mu_n \right| &\leq \sum_{w \in T_n} \left| \int_{K_w} f d\mu - \int_{K_w} f d\mu_n \right| \\ &\leq \sum_{w \in T_n} \mu_n(K_w) \max_{x, y \in K_w} |f(x) - f(y)| \\ &\leq \max\{|f(x) - f(y)| : w \in T_n, x, y \in K_w\} \rightarrow 0, \end{aligned}$$

using uniform continuity for the last inequality.

7. Replication and Trace for a Single IFS

7.1. Assumptions. Consider a *single* pcf IFS $F = (\psi_1, \dots, \psi_M)$, see Section 3.

Assume the following intersection condition:

$$(15) \quad \text{Any two distinct 1-cells } \psi_m(V^{(0)}) \text{ and } \psi_n(V^{(0)}) \text{ intersect in at most one point.}$$

The set of intersection points is the critical set B^λ from Proposition 4.1(3), and from (2) in Section 3.1. See Figure 2.

The assumption holds in the cases of interest here, although it is not known if it holds in general for nested fractals, see [Bar98, p71 (5.16)].

Let $\rho = (\rho_1, \dots, \rho_M)$ where $\rho_m > 0$. Then ρ is called a *resistance scaling vector*.

Example 7.1. For the IFS corresponding to $SG(2)$ the natural value is $\rho = (\frac{5}{3}, \frac{5}{3}, \frac{5}{3})$ and for $SG(3)$ it is $\rho = (\frac{15}{7}, \dots, \frac{15}{7})$. See Section ***.

7.2. Definition of Replication. Suppose for $1 \leq m \leq M$ that $V^{(0)} \subset W^m$ where W^m is finite and (W^m, A^m) is a network with conductance matrix $A^m = (a_{xy}^m)$. Let $\mathcal{E}_m$ be the Dirichlet form on (W^m, A^m) .

In particular, the case $W^m = V^{(0)}$ for all m is important.

Assume for $m \neq n$ that

$$(16) \quad \psi_m(W^m) \cap \psi_n(W_n) = \psi_m(V^{(0)}) \cap \psi_n(V^{(0)}).$$

Thus copies of W^m are joined together by joining the corresponding copies of $V^{(0)}$ in the standard manner to form $V^{(1)}$, but no other nodes are joined. This is true, for example, if $W^m \subset E$, where E is the set used in the Definition from Section 3.1 after replacing F^λ by F .

Informally, the *replication network* (W^R, A^R) using (F, ρ) is obtained by multiplying the conductances a_{xy}^m by ρ_m , applying ψ_m to W^m , and joining the resulting networks together at their common vertices (which form the critical subset of $V^{(1)}$). Note that

$$(17) \quad V^{(0)} \subset W^m \quad \forall m, \quad V^{(0)} \subset V^{(1)} \subset W^R.$$

The corresponding Dirichlet form on (W^R, A^R) is denoted by $\mathcal{E}^R = \mathcal{E}_A^R$.

More precisely:

Definition 7.2. The *replication network* (W^R, A^R) using (F, ρ) is given by

$$(18) \quad \begin{aligned} W^R &= \bigcup_{m=1}^M \psi_m(W^m), \\ a_{xy}^R &= \begin{cases} \rho_m a_{vw}^m & \text{if } x \neq y, \ v, w \in V^{(0)}, \ \psi_m(v) = x, \ \psi_m(w) = y \\ 0 & \text{otherwise,} \end{cases} \\ a_{xx}^R &= \sum \{\rho_m a_{vv}^m : \psi_m(v) = x\}. \end{aligned}$$

Concerning the definition of a_{xy}^R , note that by assumption there is at most one m and one pair v, w satisfying the condition used in the definition of a_{xy}^R . Concerning a_{xx}^R , there will be more than one m and v precisely when x is in the critical set.

7.3. Invariance under Replication.

Definition 7.3. Suppose $(V^{(0)}, A)$ is a network on $V^{(0)}$ with replication network $(V^{(0)}, A^R)$ under (F, ρ) as in Section 7.2. Then $(V^{(0)}, A)$ is *invariant under replication* if⁶

$$A = \text{Tr}(A^R \mid V^{(0)}).$$

Example 7.1 gives an example of an invariant network if $B = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$.

7.4. Properties of Replication. Suppose a network is obtained by replication as in the previous section, with resistance scalings ρ_m . The following Proposition shows the following:

- (1) The Dirichlet energy of a function f defined on the replicated network can be obtained from the Dirichlet energy of the corresponding function on each of the original networks by scaling these energies with ρ_m and summing.
- (2) The trace of the replicated network on $V^{(1)}$ is the same as the replication of the traces on $V^{(0)}$ of the original networks.

Proposition 7.4. Suppose $F = (\psi_1, \dots, \psi_m)$ is a pcf fractal satisfying the intersection condition (15), and $\rho = (\rho_1, \dots, \rho_M)$ is a resistance scaling vector.

Suppose (W^m, A^m) is a network with conductance matrix $A^m = (a_{xy}^m)$ for $1 \leq m \leq M$. Suppose $V^{(0)} \subset W^m$ for all m and $\psi_m(W^m) \cap \psi_n(W_n) = \psi_m(V^{(0)}) \cap \psi_n(V^{(0)})$ for $m \neq n$.

Let the replication network (W^R, A^R) be defined by (18).

⁶For trace networks see [Hut09].

Then the matrix A^R is a conductance matrix. The Dirichlet form $\mathcal{E}_A^R$ is given by

$$(19) \quad \mathcal{E}_A^R(f, g) = \frac{1}{2} \sum_{x, y} a_{xy}^R (f(x) - f(y))(g(x) - g(y)) = \sum_{m=1}^M \rho_m \mathcal{E}_m(f \circ \psi_m, g \circ \psi_m).$$

Suppose $B^m = \text{Tr}(A^m \mid V^{(0)})$. Let $(V^{(1)}, B^R)$ be the replication network obtained by applying ψ_m and ρ_m to $(V^{(0)}, B^m)$ for $1 \leq m \leq M$. Then $B^R = \text{Tr}(A^R \mid V^{(1)})$.

Proof. It is clear $a_{xy}^R \geq 0$ for $x \neq y$. Moreover,

$$\sum_{\{y: y \neq x\}} a_{xy}^R = \sum_{\{m: x \in V_m^{(0)}\}} \sum_{\{y: y \in V_m^{(0)}, y \neq x\}} a_{xy}^R$$

(each $y \neq x$ is counted once on the right side by (16) and the initial assumption about the intersection of 1-cells)

$$\begin{aligned} &= \sum_{\{m: x \in V_m^{(0)}\}} \sum_{\{w: w \in V_m^{(0)}, w \neq \psi_m^{-1}(x)\}} \rho_m a_{\psi_m^{-1}(x)w} \\ &= - \sum_{\{m: x \in V_m^{(0)}\}} \rho_m a_{\psi_m^{-1}(x) \psi_m^{-1}(x)} \\ &= -a_{xx}^R. \end{aligned}$$

Hence A^R is a conductance matrix.

The first equality in (19) is by definition. The second is by splitting the first sum into disjoint sums over the sets $V_m^{(0)}$.

For the final claim it is sufficient to show that if $g: V^{(1)} \rightarrow \mathbb{R}$ and f is its harmonic extension to (W^R, A^R) then

$$\mathcal{E}_B^R(g) = \mathcal{E}_A^R(f).$$

But f is the harmonic extension of g to (W^R, A^R) iff, for every m , $f \circ \psi_m$ is the harmonic extension of $g \circ \psi_m: V^{(0)} \rightarrow \mathbb{R}$ to (W^m, A^m) . To see this, consider $x \in W \setminus V^{(1)}$ and let m be the unique integer such that $x \in \psi_m(W^m)$, see (16). Then g is harmonic at x iff $g \circ \psi_m$ is harmonic at $\psi_m^{-1}(x)$, because all relevant conductances are scaled by the same factor ρ_m . $\square$

Corollary 7.5. *Continue the assumptions of Proposition 7.4. Suppose $(V^{(0)}, B)$ is invariant under replication and $\text{Tr}(A^m \mid V^{(0)}) = B$ for all m .*

Then $\text{Tr}(A^R \mid V^{(0)}) = B$.

Proof.

$$\begin{aligned} \text{Tr}(A^R \mid V^{(1)}) &= B^R \quad \text{by Proposition 7.4,} \\ \text{Tr}(B^R \mid V^{(0)}) &= B \quad \text{by assumption,} \\ \therefore \text{Tr}(A^R \mid V^{(0)}) &= B \quad \text{by the tower property of traces.} \end{aligned} \quad \square$$

REFERENCES

- [Bar98] Martin T. Barlow, *Diffusions on fractals*, Lectures on probability theory and statistics (Saint-Flour, 1995), Lecture Notes in Math., vol. 1690, Springer, Berlin, 1998, pp. 1–121.
- [Ham97] B. M. Hambly, *Brownian motion on a random recursive Sierpinski gasket*, Ann. Probab. **25** (1997), 1059–1102.
- [Hut09] J. E. Hutchinson, *Electric Currents, Random Walks and Analysis on Graphs* (2009). Informal Notes.
- [Kig01] Jun Kigami, *Analysis on fractals*, Cambridge Tracts in Mathematics, vol. 143, Cambridge University Press, 2001.